

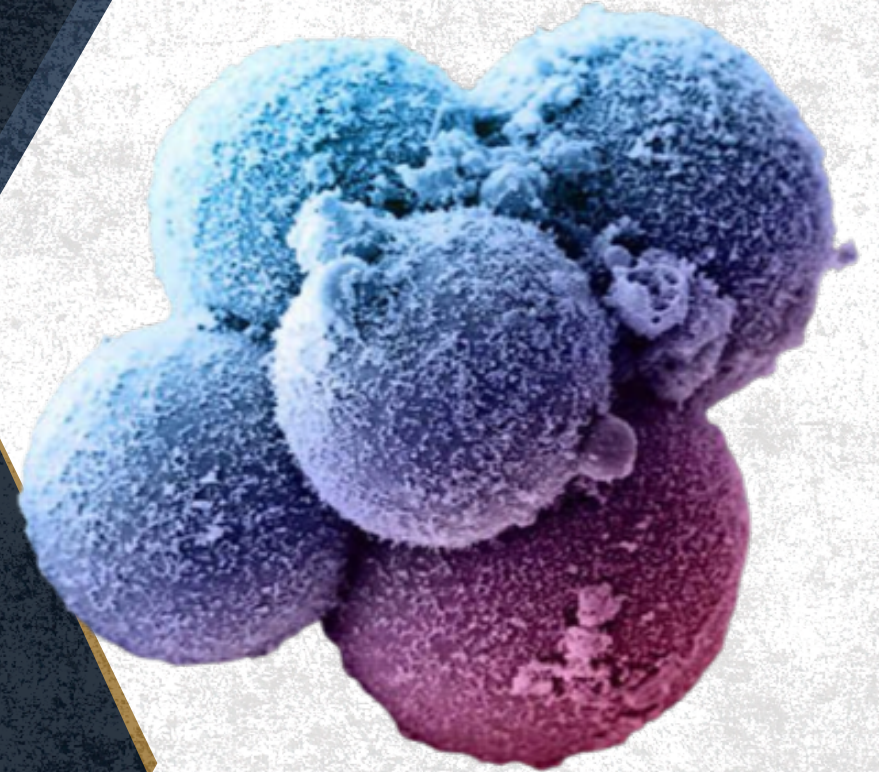


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— Presents

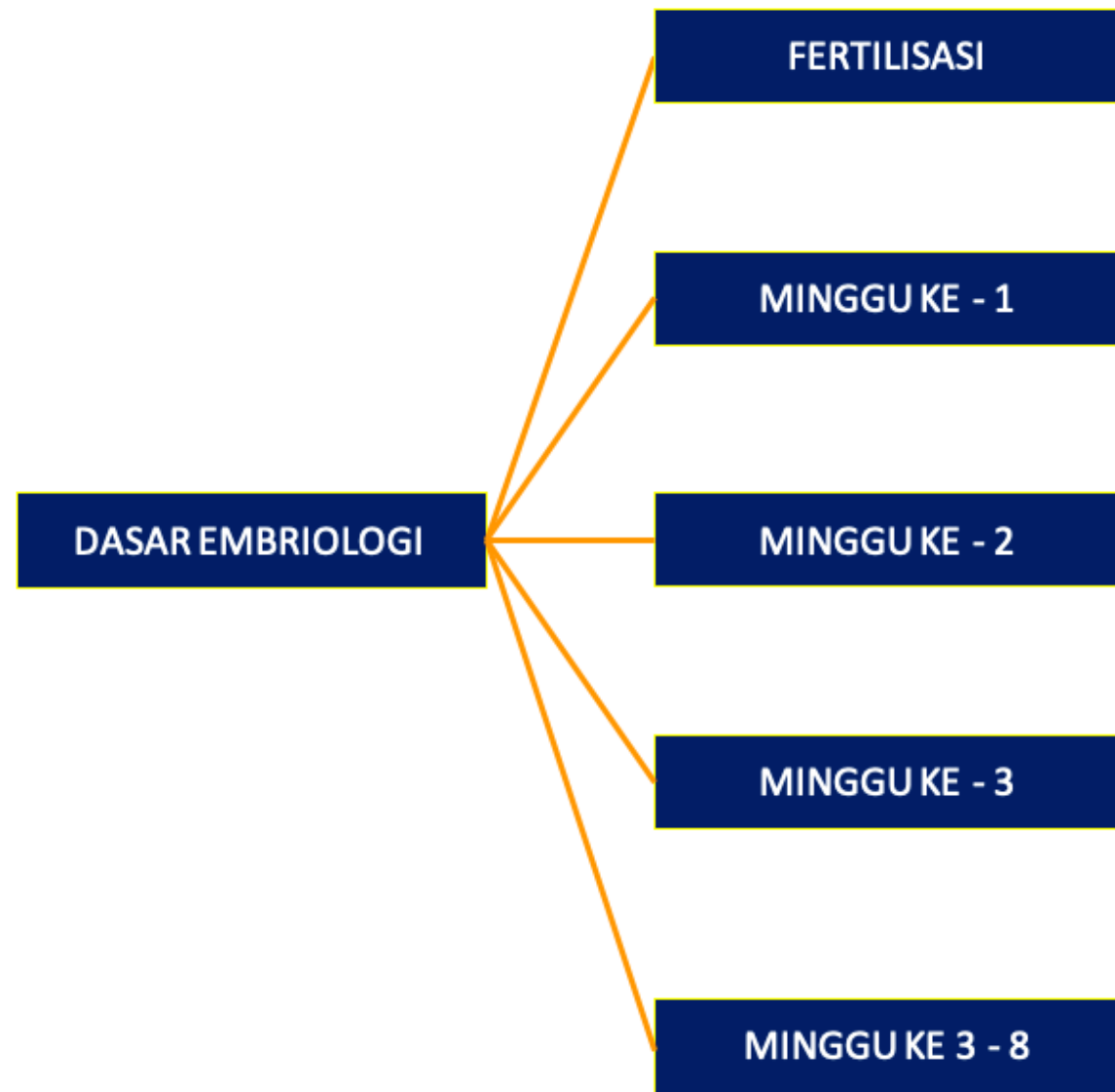
BASIC EMBRYOLOGY

M. Zydane Tsaqiifindra D. - G0024091





MIND MAP



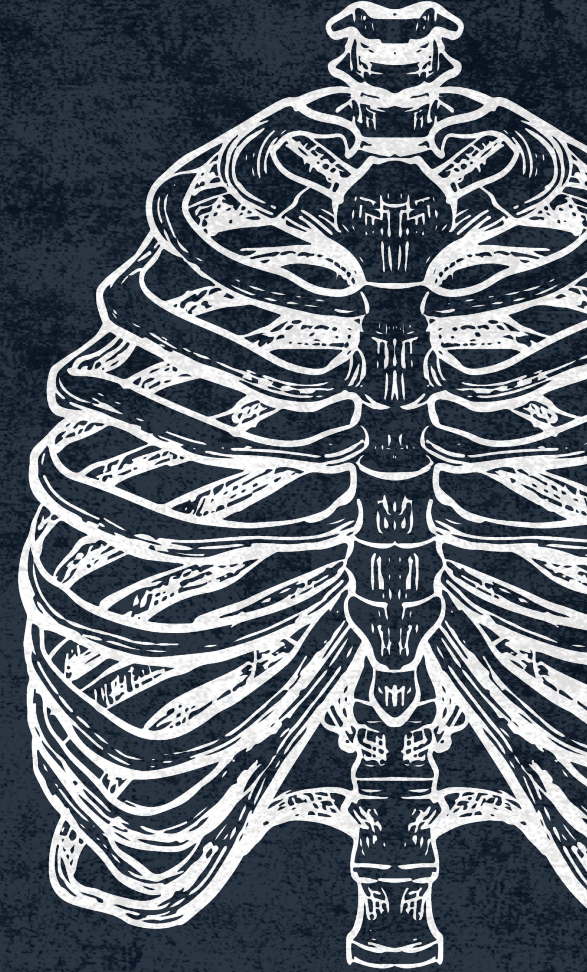


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FERTILIZATION



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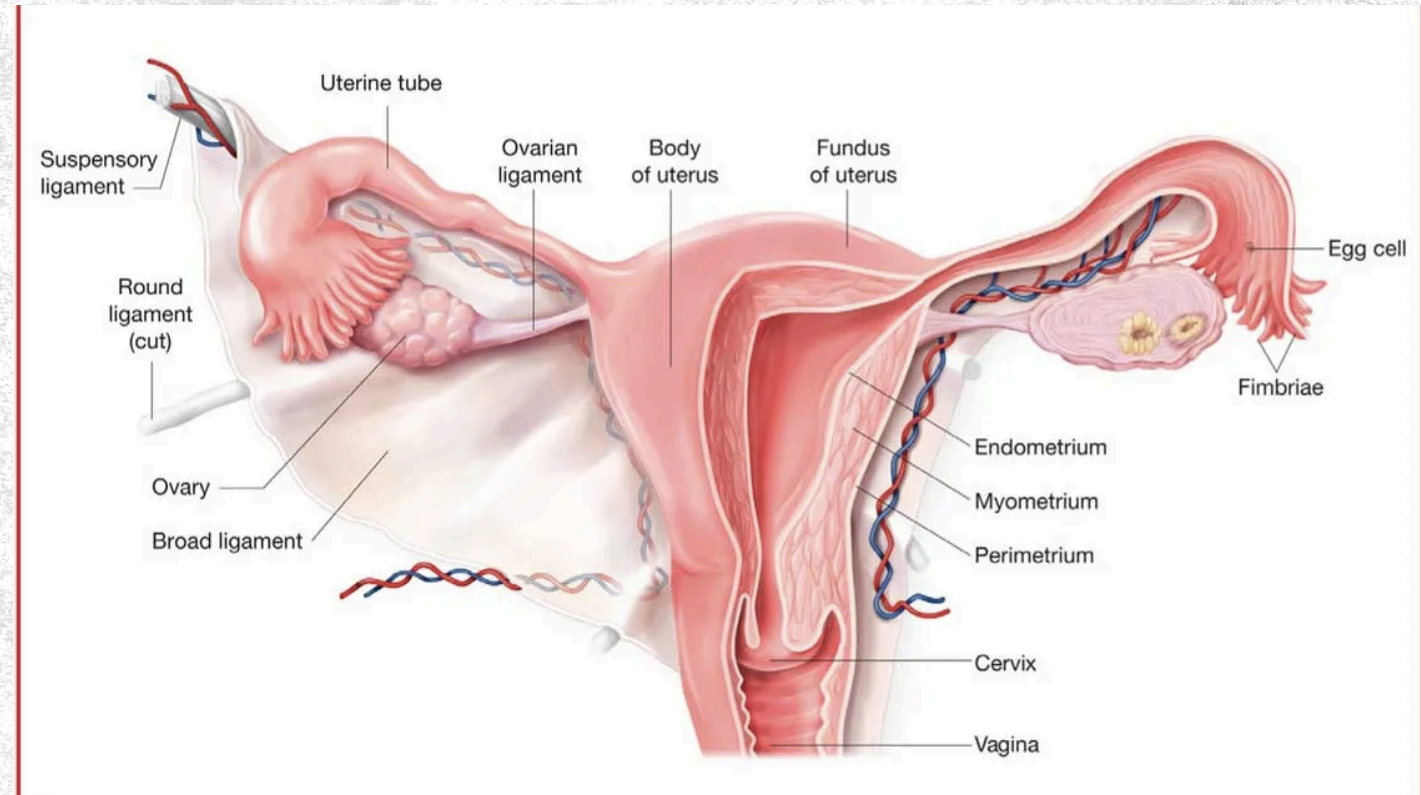
ACROMISCHE





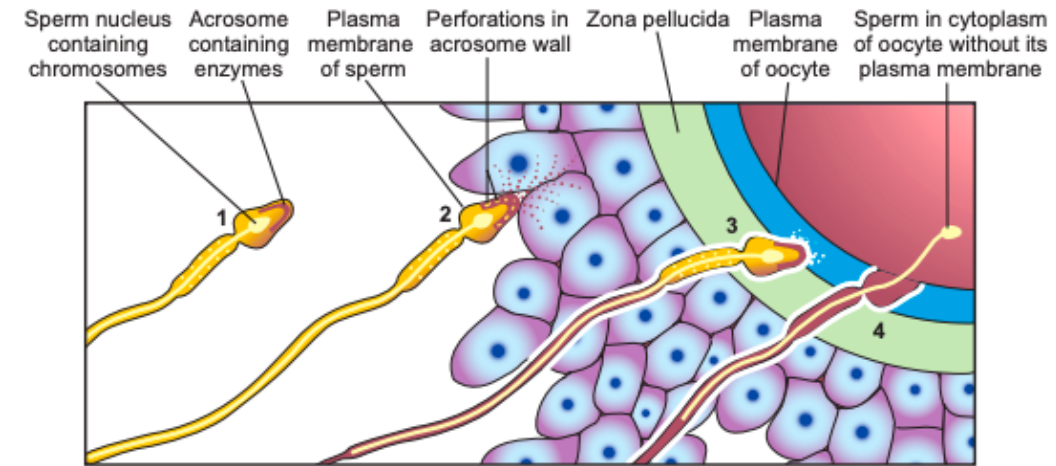
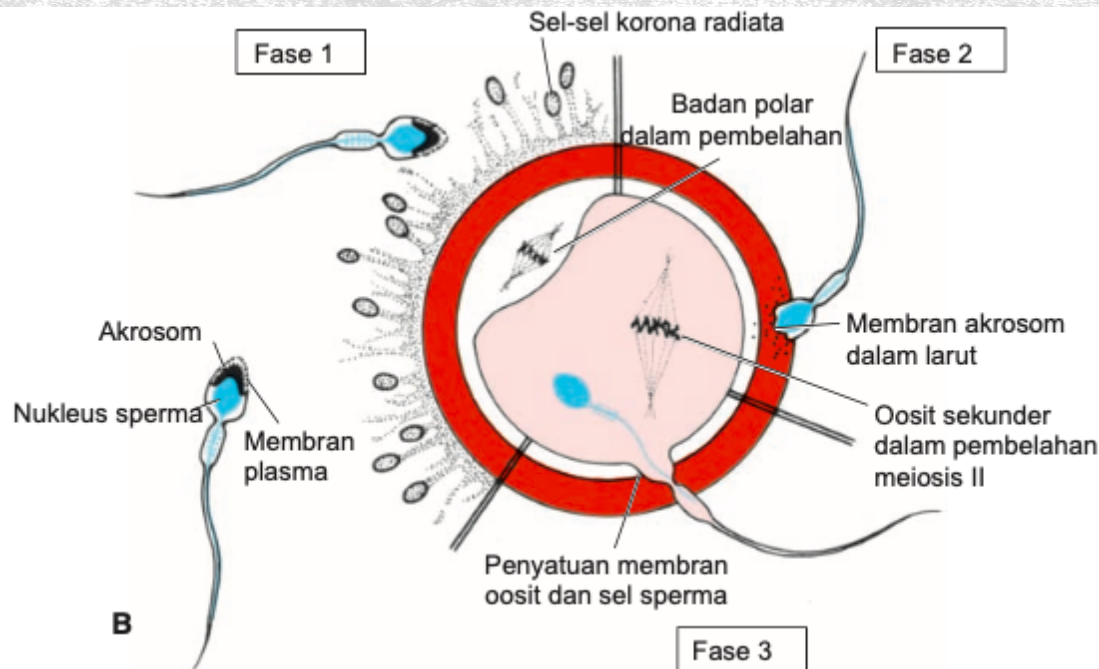
FERTILIZATION

To form a human, the first step is fertilization. Fertilization is the **meeting of ovum and sperm**. The meeting is usually occur in tuba uterine.





FERTILIZATION

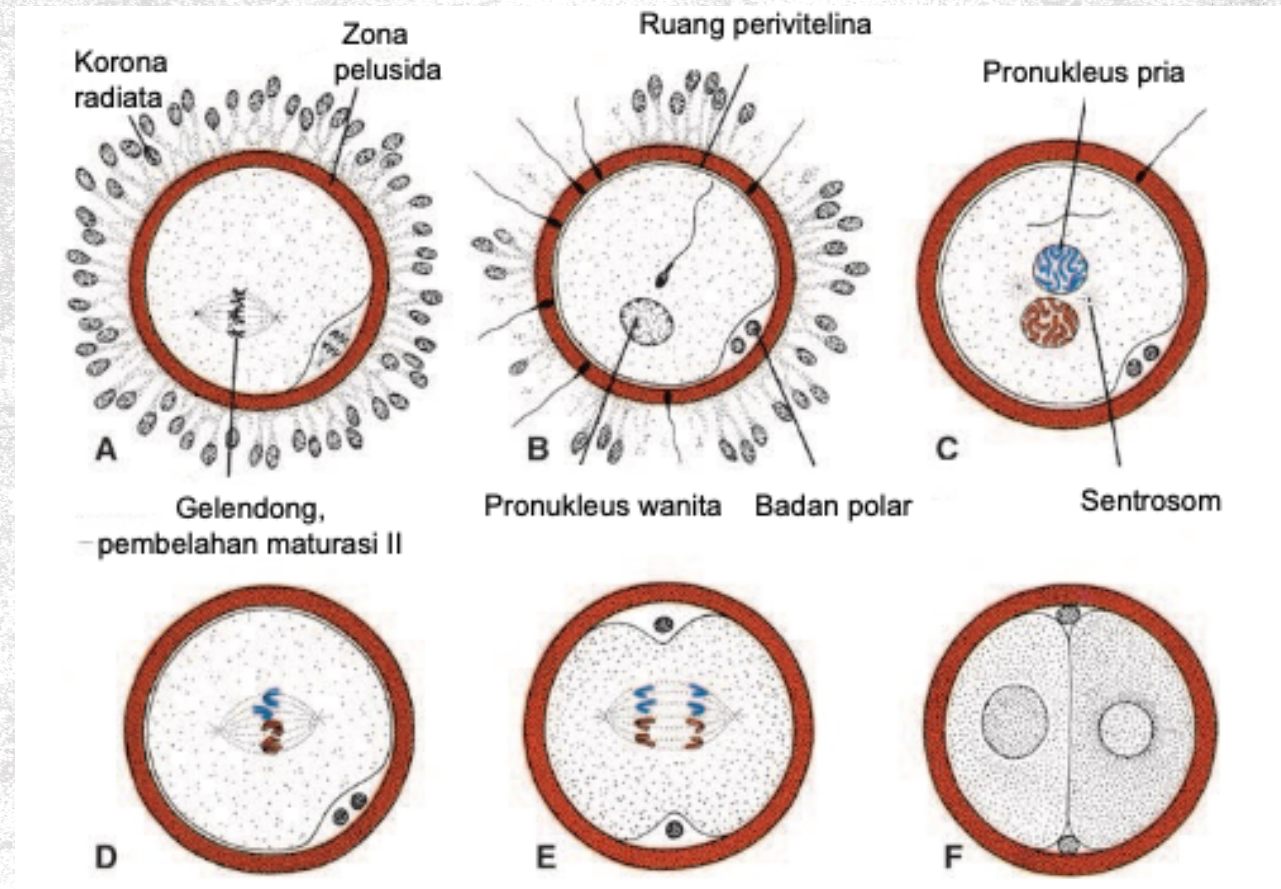


CAPACITATION → ACROSOMAL REACTION → FERTILIZATION → LYSOZIME RELEASE





FERTILIZATION



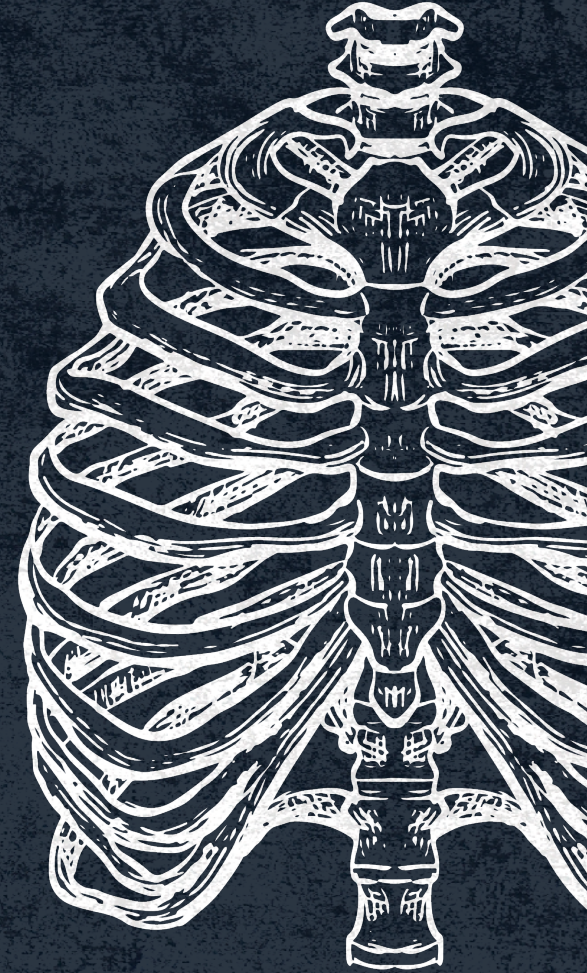


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FIRST WEEK



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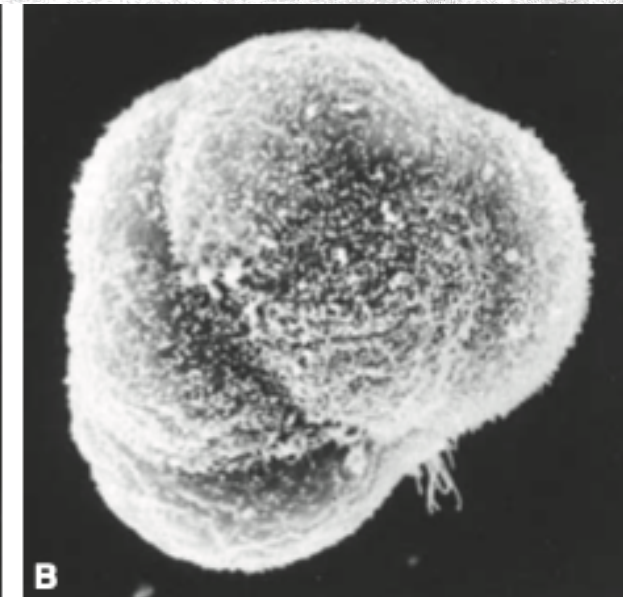
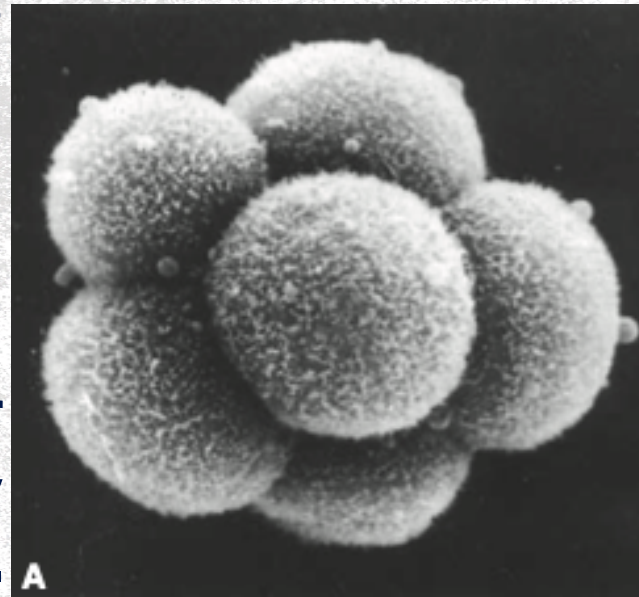




FIRST WEEK

1. Blastulasi

After divided into 2 section it called cleavage, zygote will divided until 8 section. This 8 section is called **Blastomer**. Blastomer will unite and solidify and form **Morula/16-section**

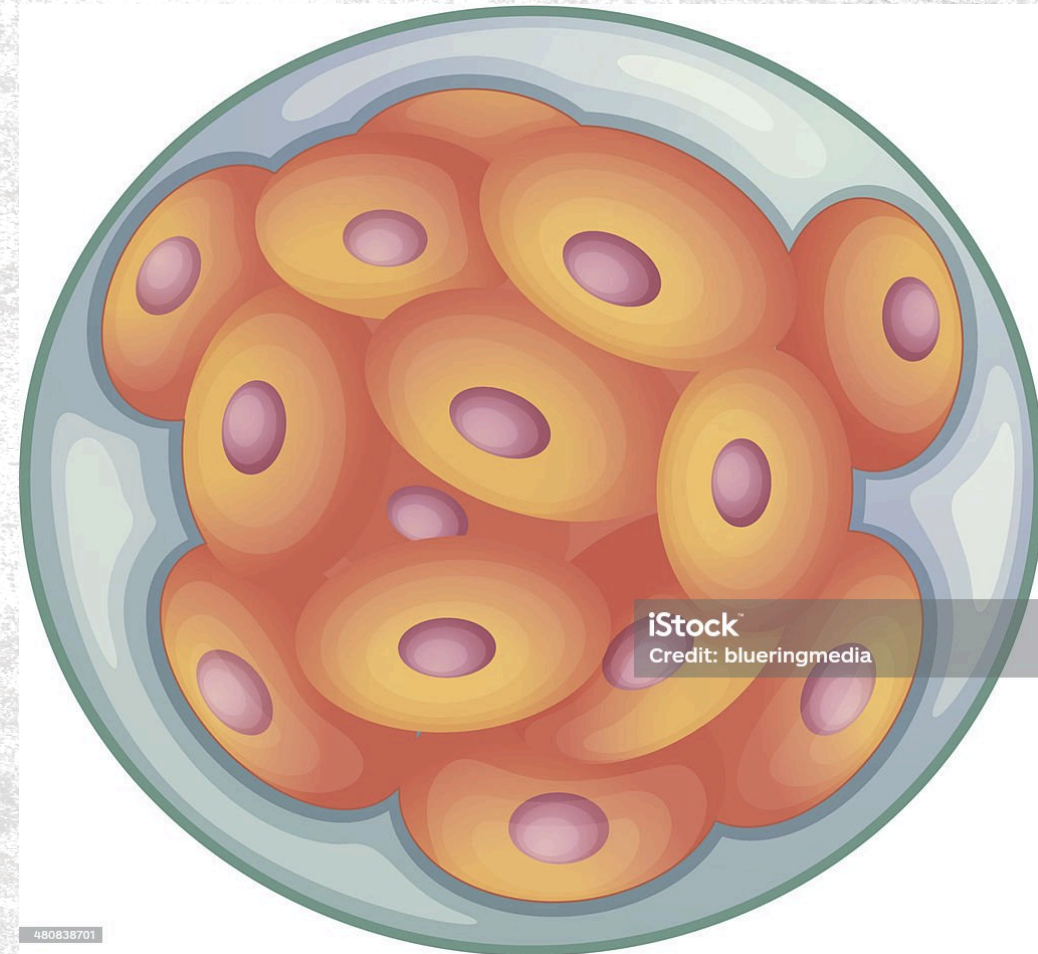




FIRST WEEK

2. Morula

16-Section/morula will divided into **inner cell mass** and **outer cell mass**. Inner cell mass wil form embryoblast, Outer cell mass will form trophoblast.

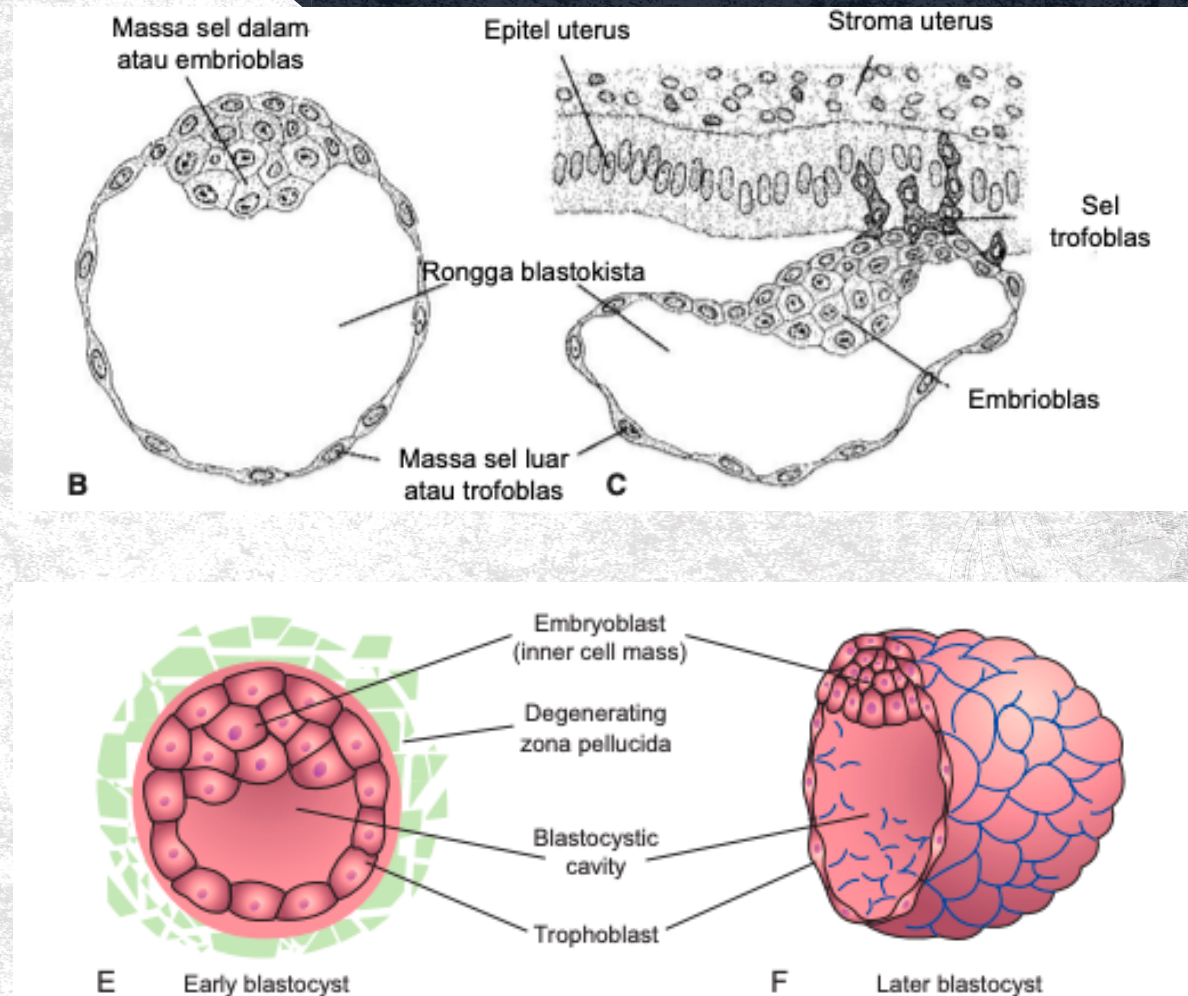




FIRST WEEK

2. Blastocyst

There will be a cavity inside morula, called blastocel, This morula is now called **blastocyst**. Blastocyst will grow and move to uterus, in day-8, blastocyst will be planted in endometrium.



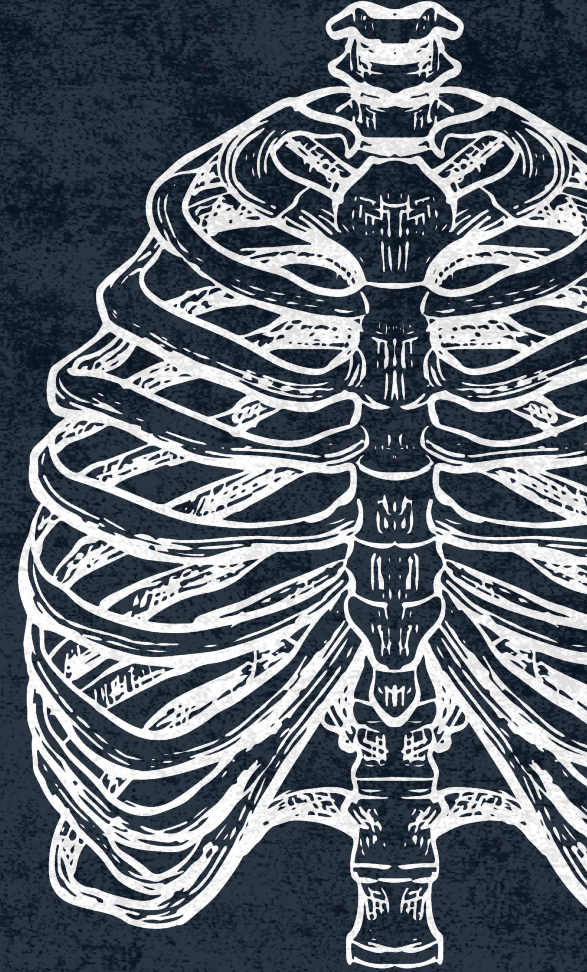


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SECOND WEEK



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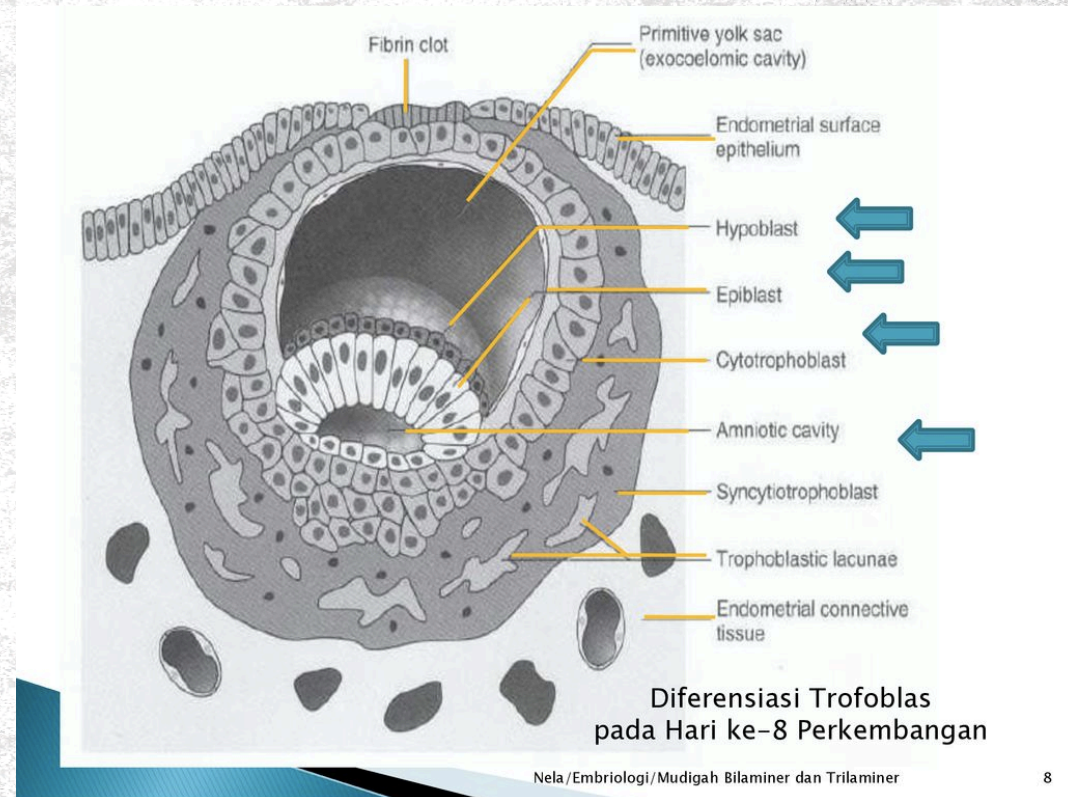




SECOND WEEK

Blastocyst will be planted successfully in endometrium of uterus.

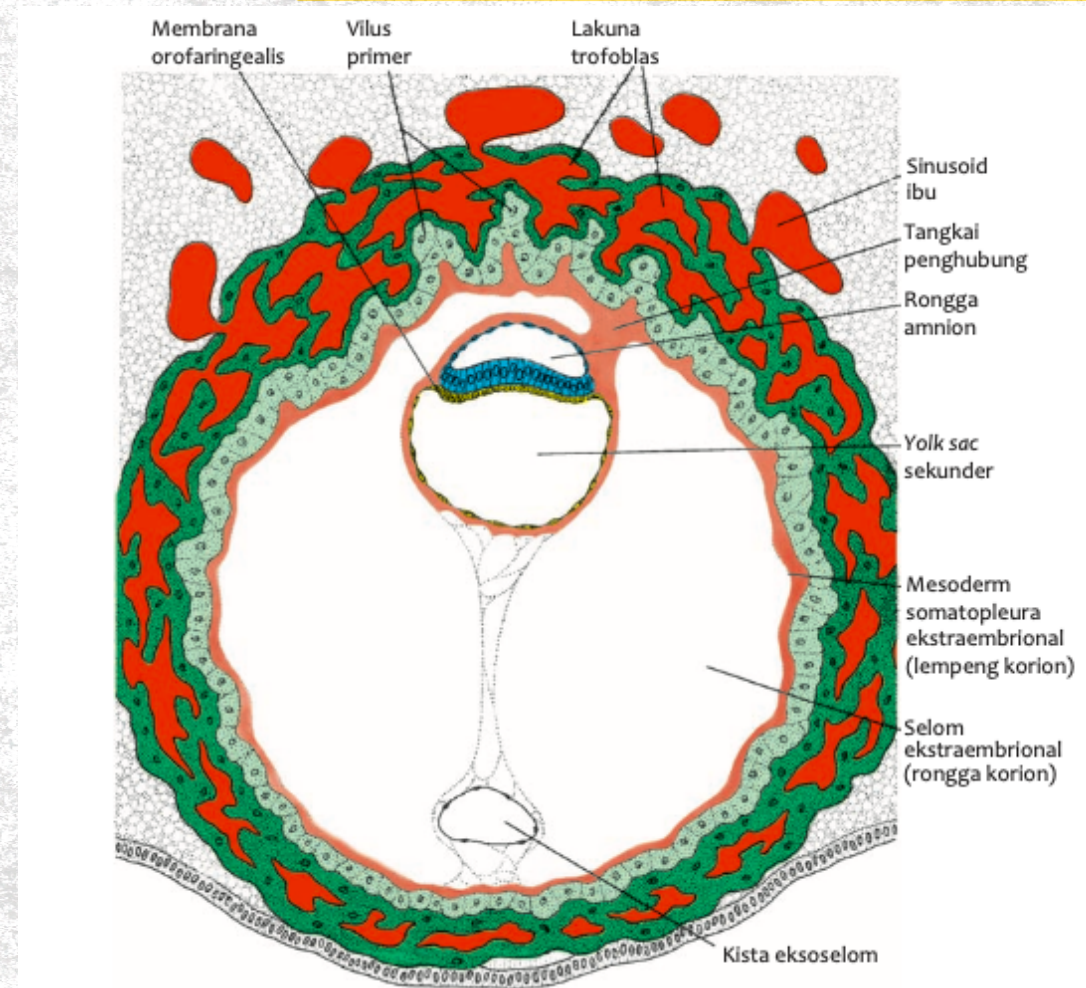
1. Embryoblast → Hypoblast
(Cuboid) & Epiblast (Cylinder)
2. Trophoblast
Sitotrophoblast (inner) and
Sinsitiotrophoblast (outer)





SECOND WEEK

Hipoblast will make a new cavity, called eksoselom cavity/primitive yolk sac.



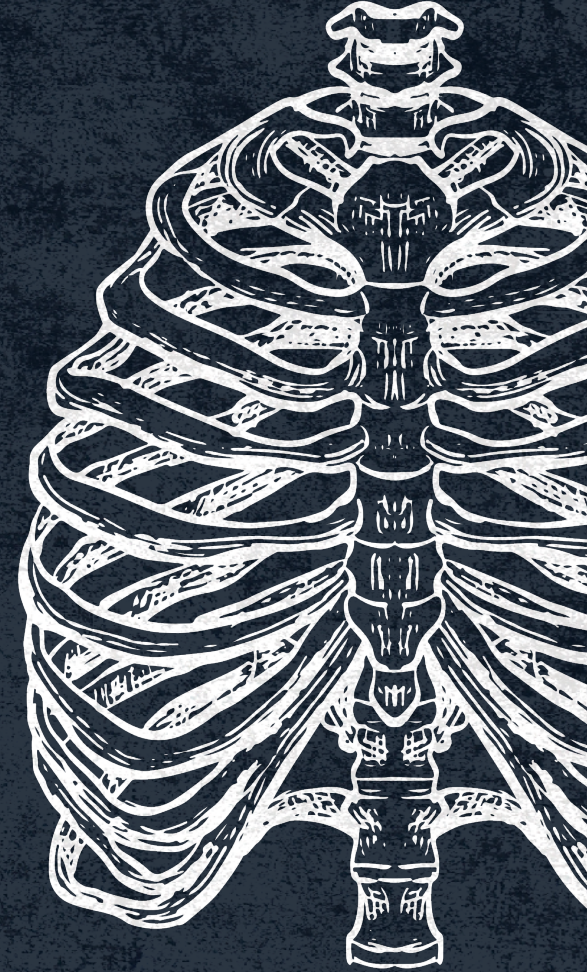


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THIRD WEEK



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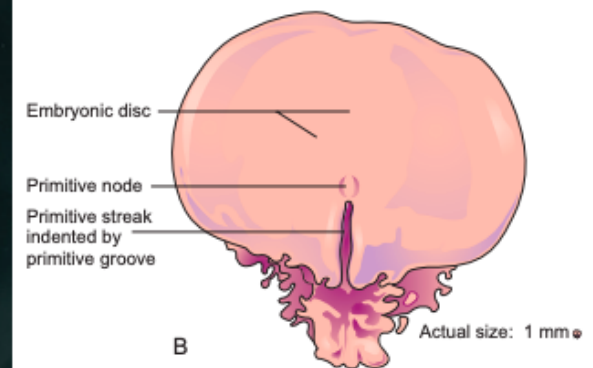
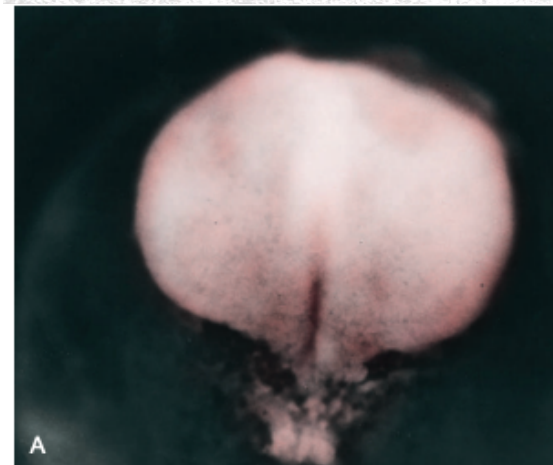
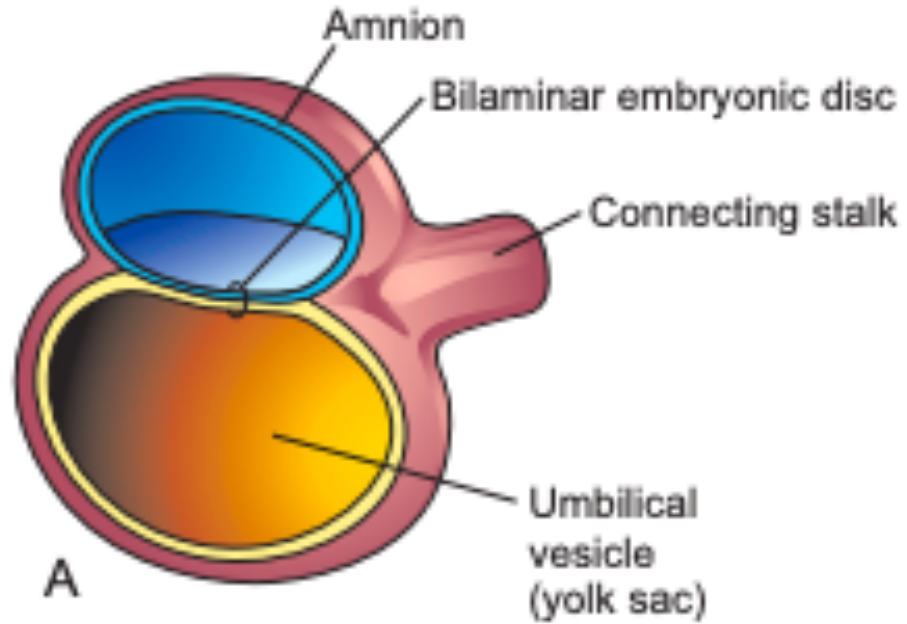
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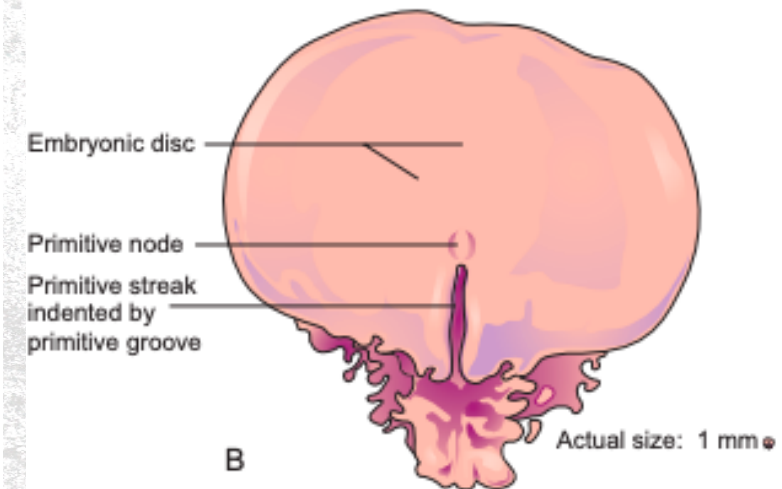
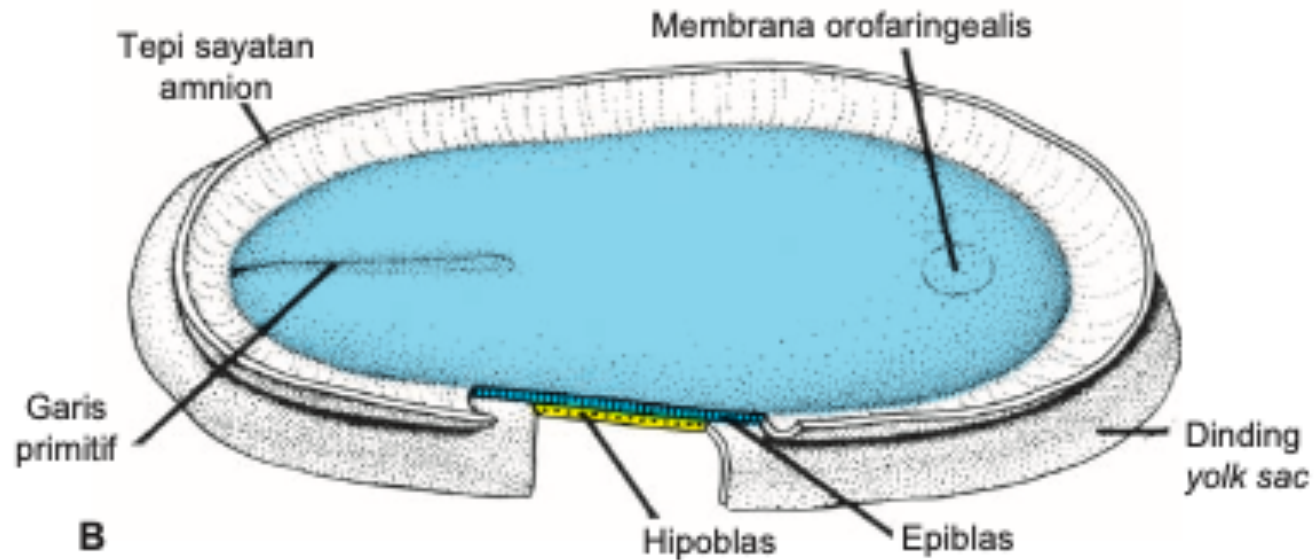


THIRD WEEK



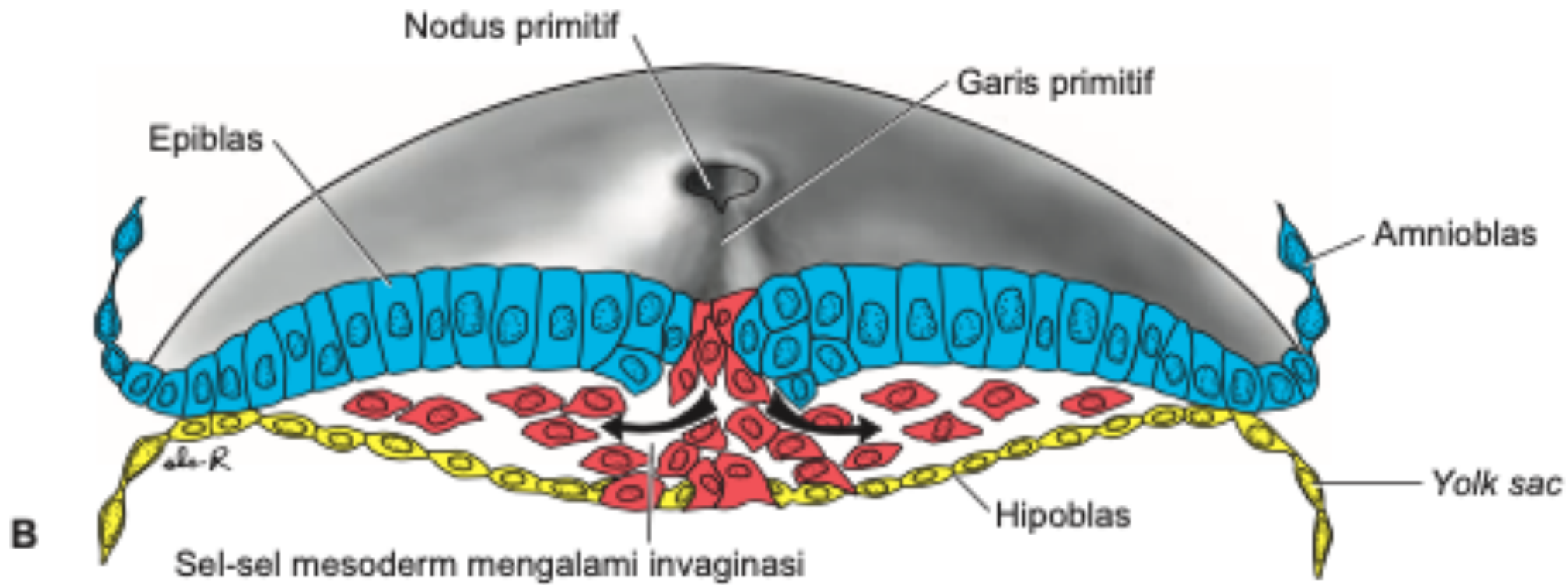


GASTRULATION



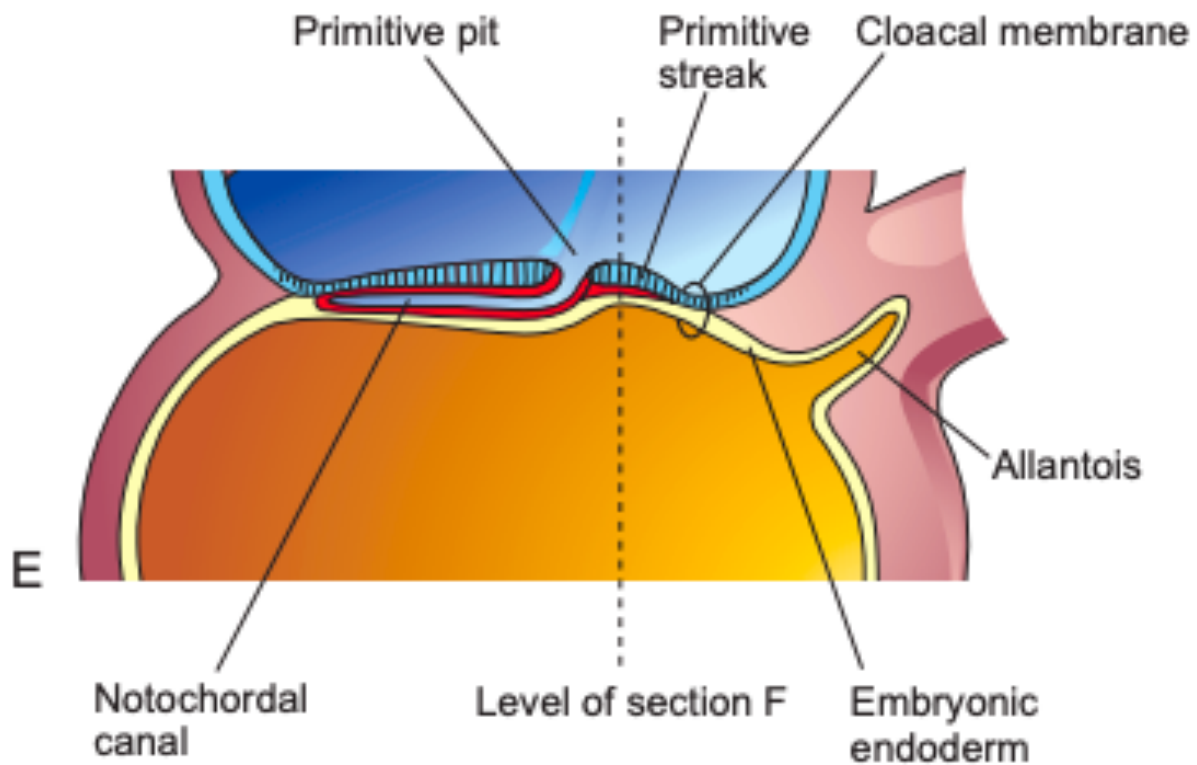


GASTRULATION

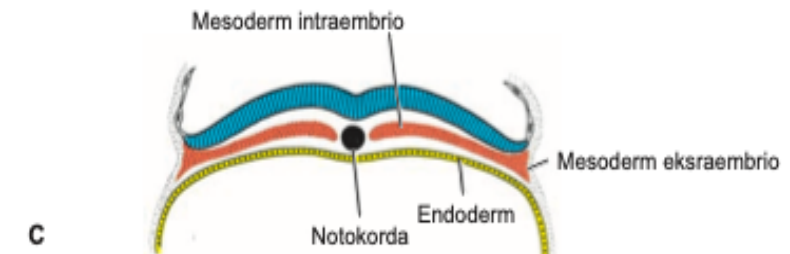




NOTOCHORD



**Notochord → Sonic
Hedgehog Protein (SHH) →
Neurulation
Notochord → Precursor of
Intervertebral Disc.**



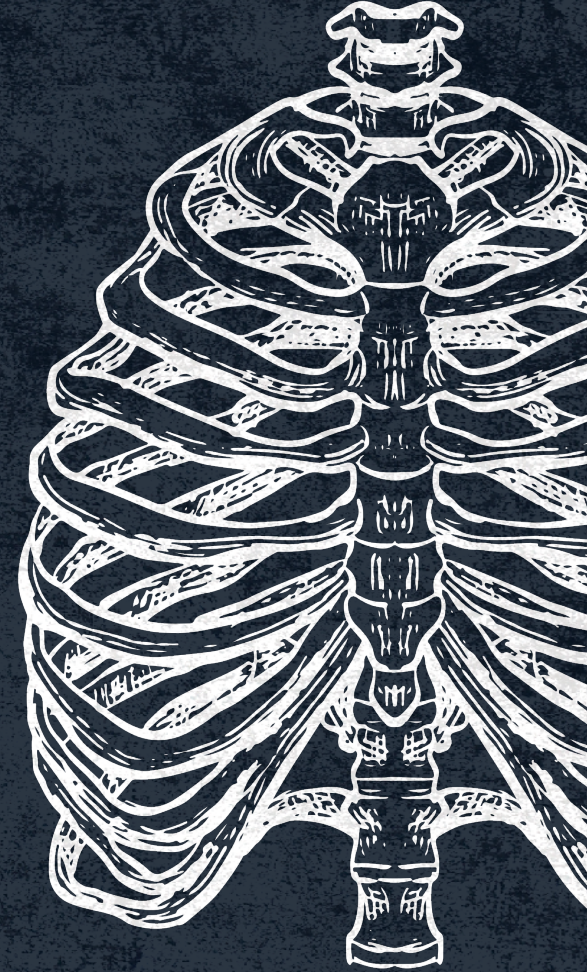


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ECTODERM



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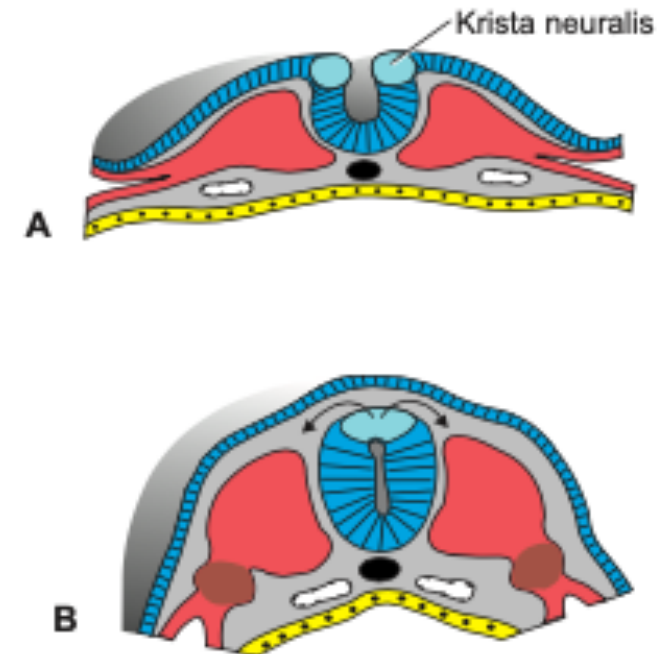
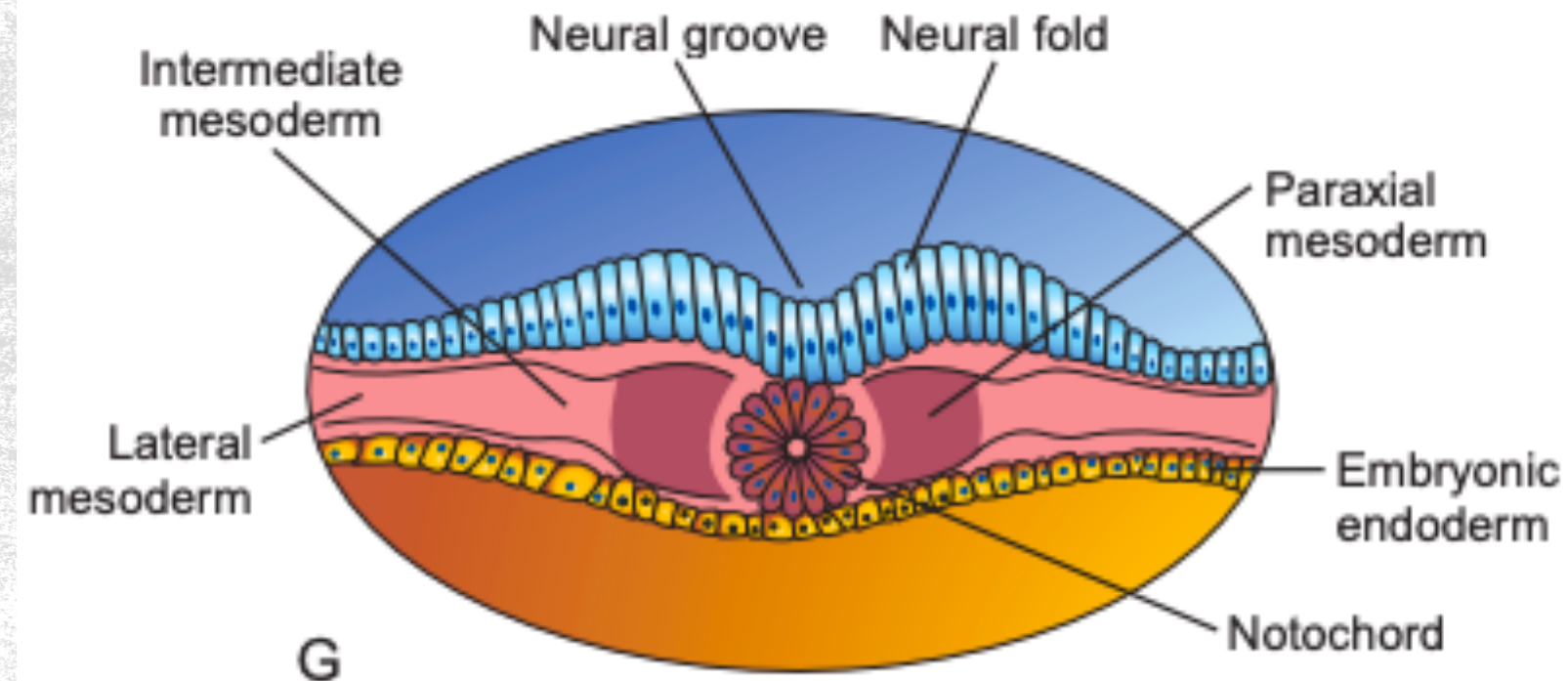
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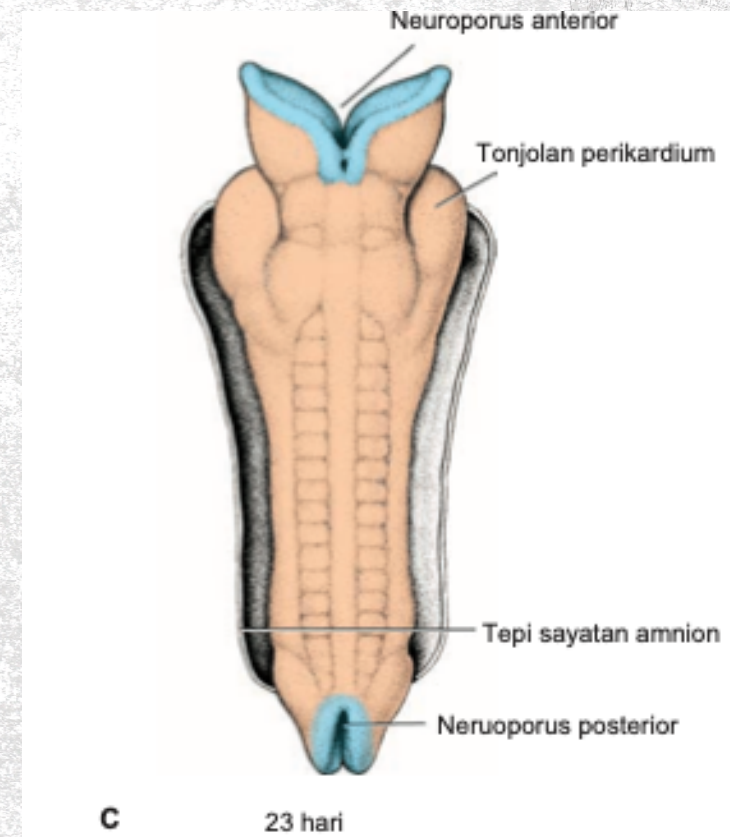
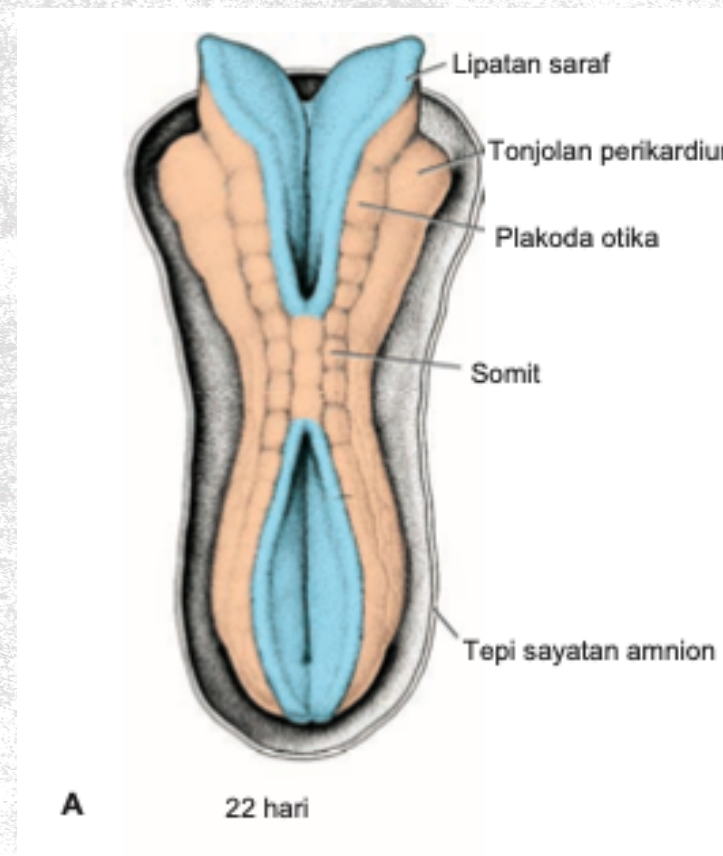
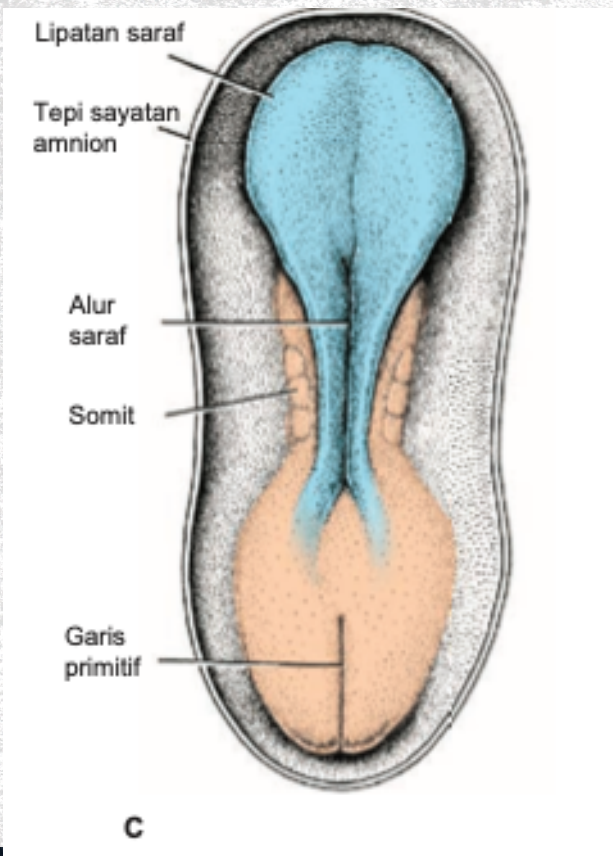


NEURULATION





NEURULATION





WHAT IF NEURAL FOLDING DOESN'T GO WELL?



Deficiency of Folic Acid (B9
Vitamin) in 3- 8 weeks of
pregnancy

NEURAL TUBE DEFECT (NTD)



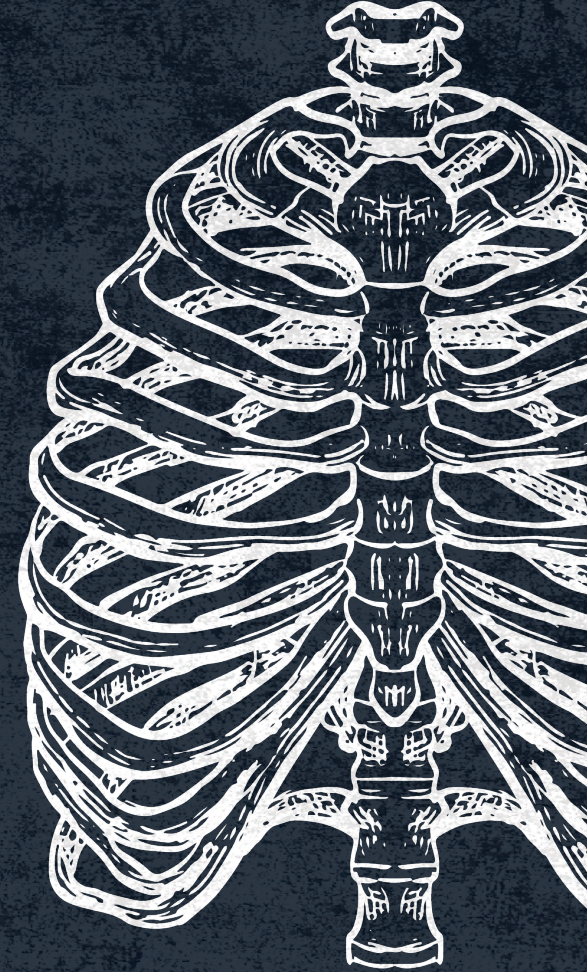


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MESODERM



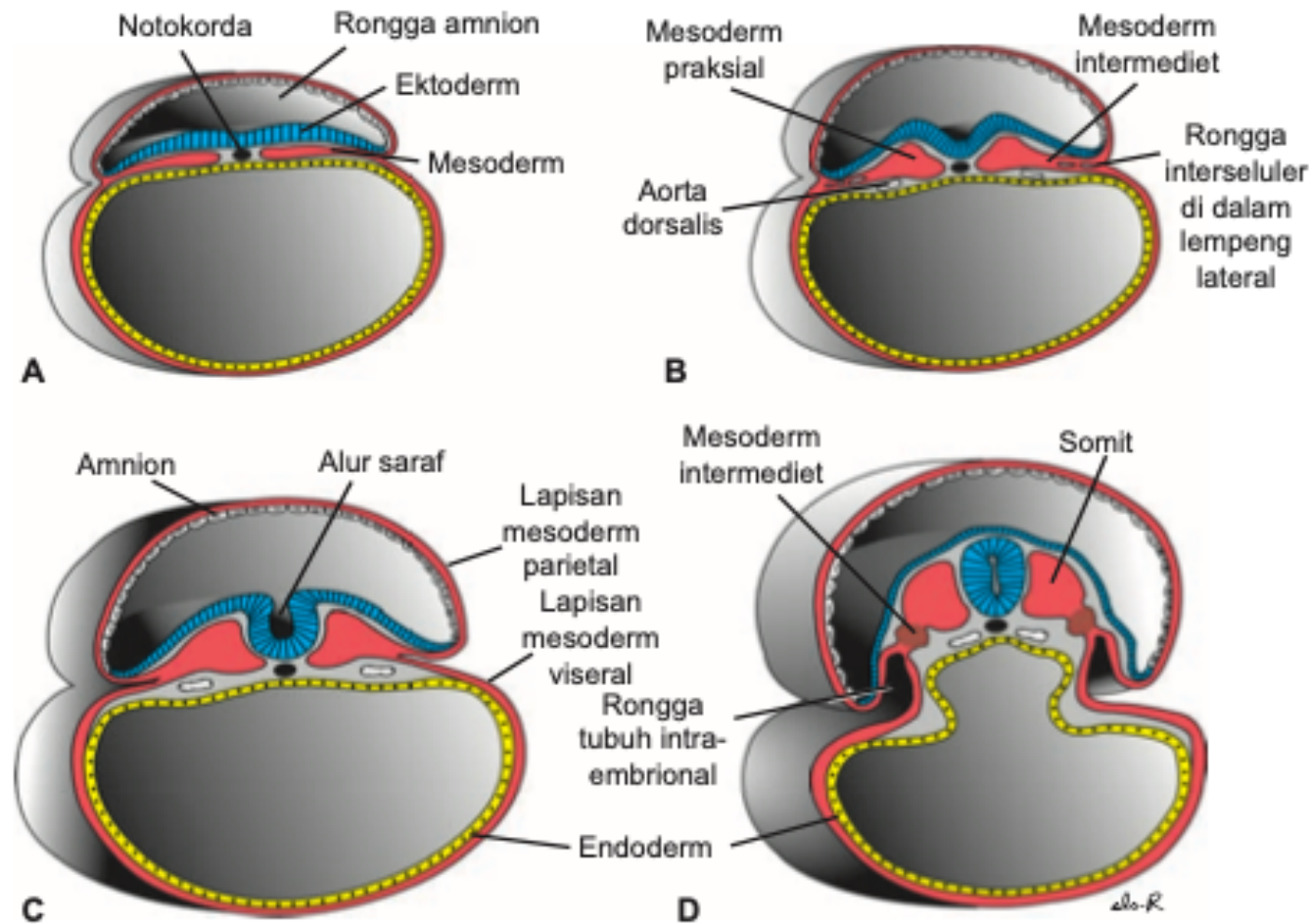
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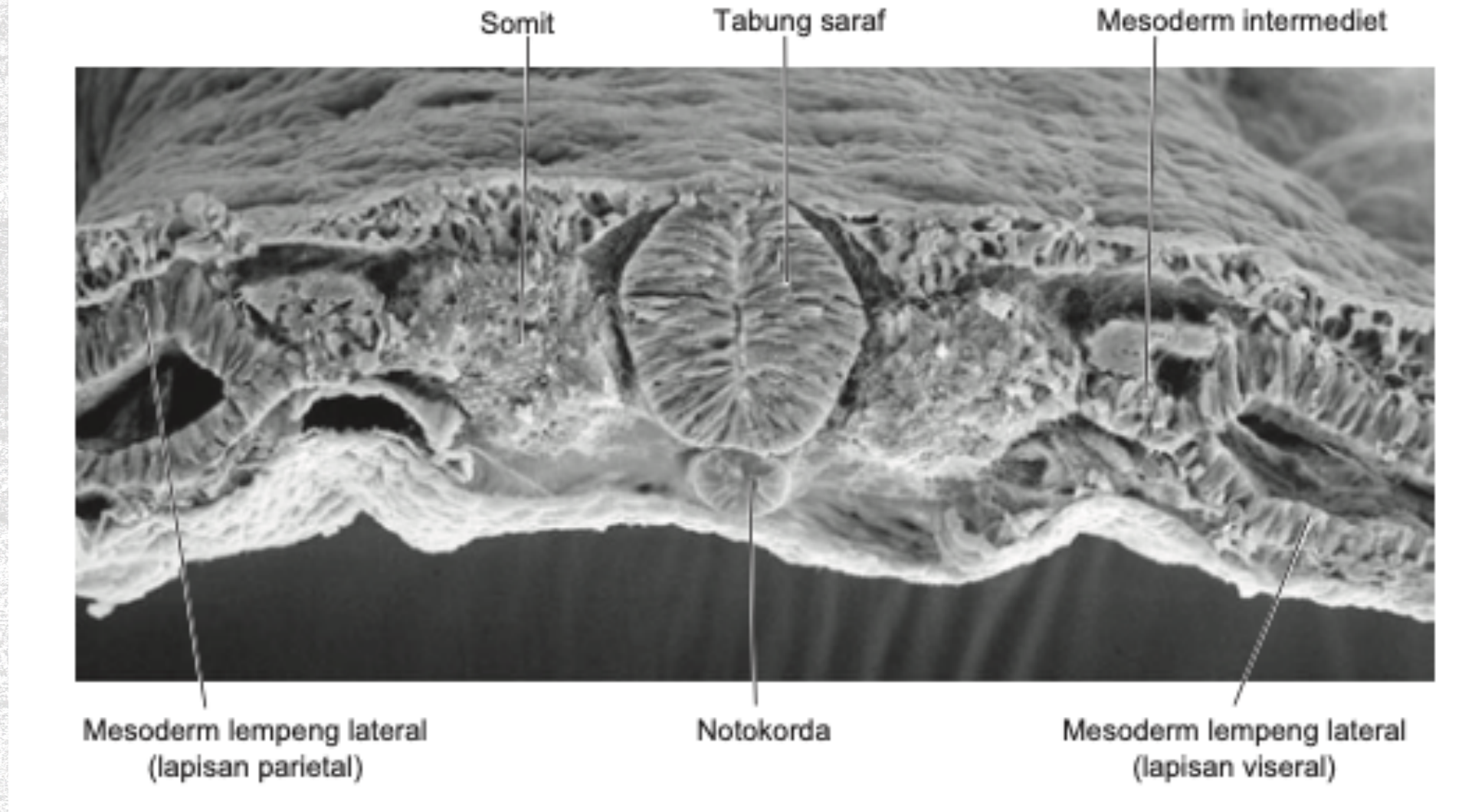
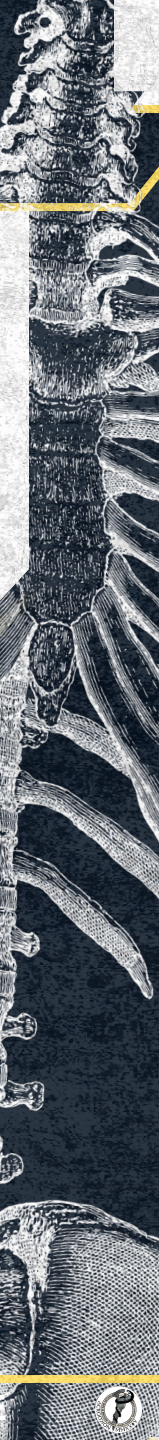


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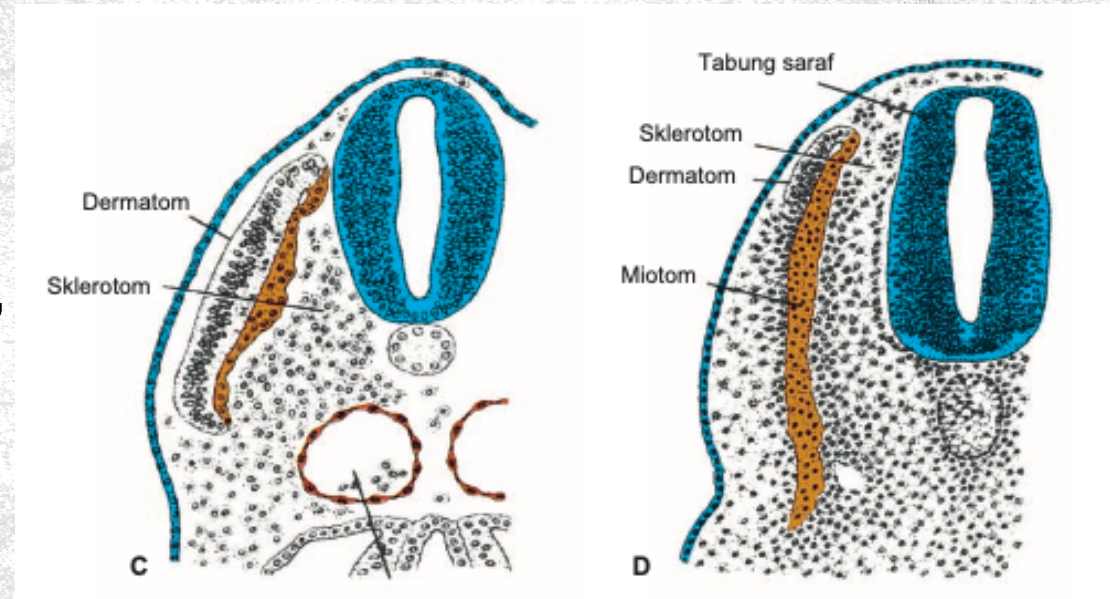
MESODERM PARAXIAL

Mesoderm Paraxial will
form:

Sclerotome will form cartilage and bones,
especially vertebrae

Dermatome will form dermis

Miotome will form muscle





MESODERM INTERMEDIET

Intermediet Mesoderm will form gonad, testicle, ovary, ren, etc



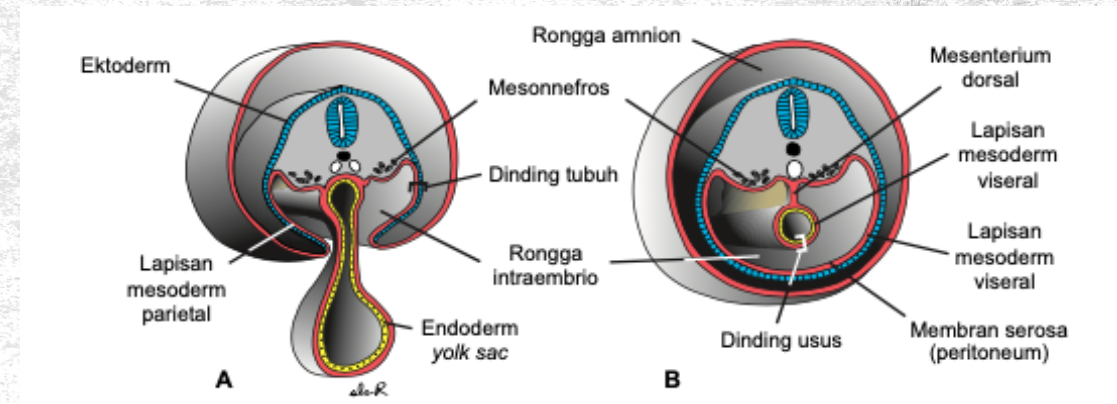


MESODERM LATERAL

Lateral Mesoderm consist of:
somatic layer of lateral mesoderm and **splachic layer of lateral mesoderm**.

Splachic layer: Heart, Lungs, Adrenal gland, and many other visceral organ

Somatic layer: Upper and lower extremity



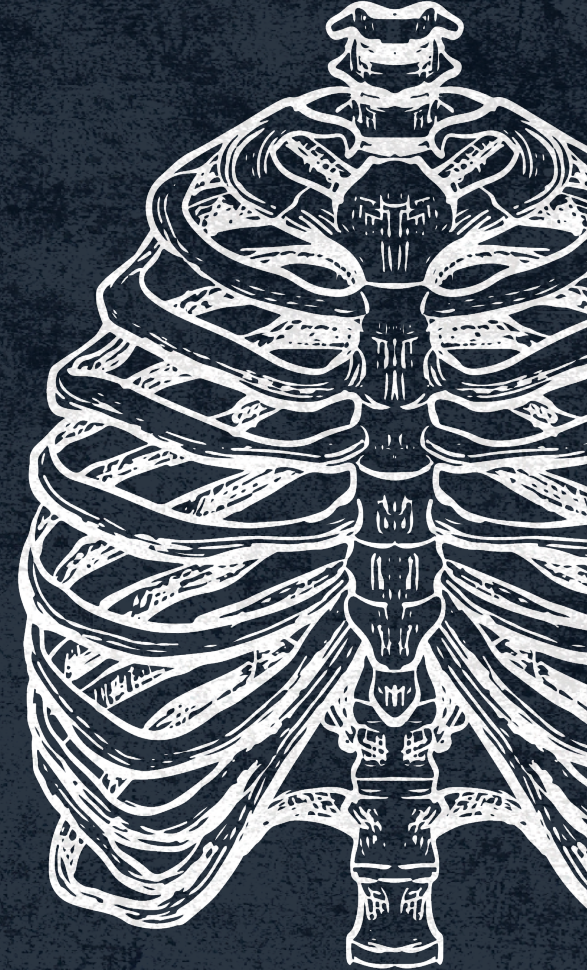


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ENDODERM



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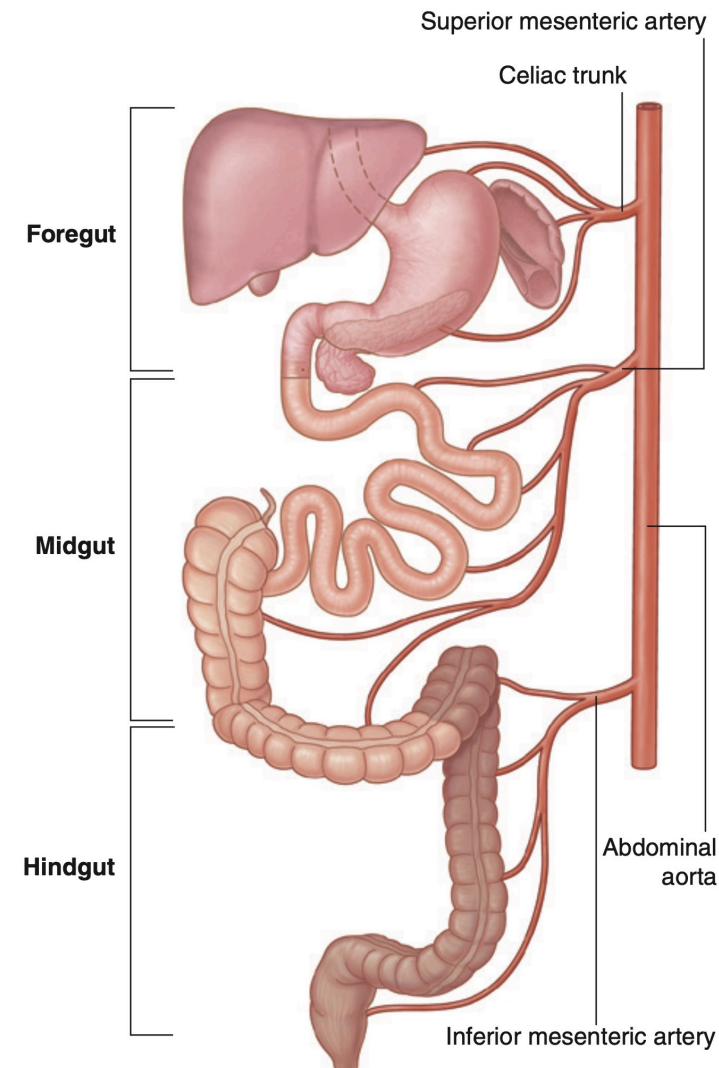
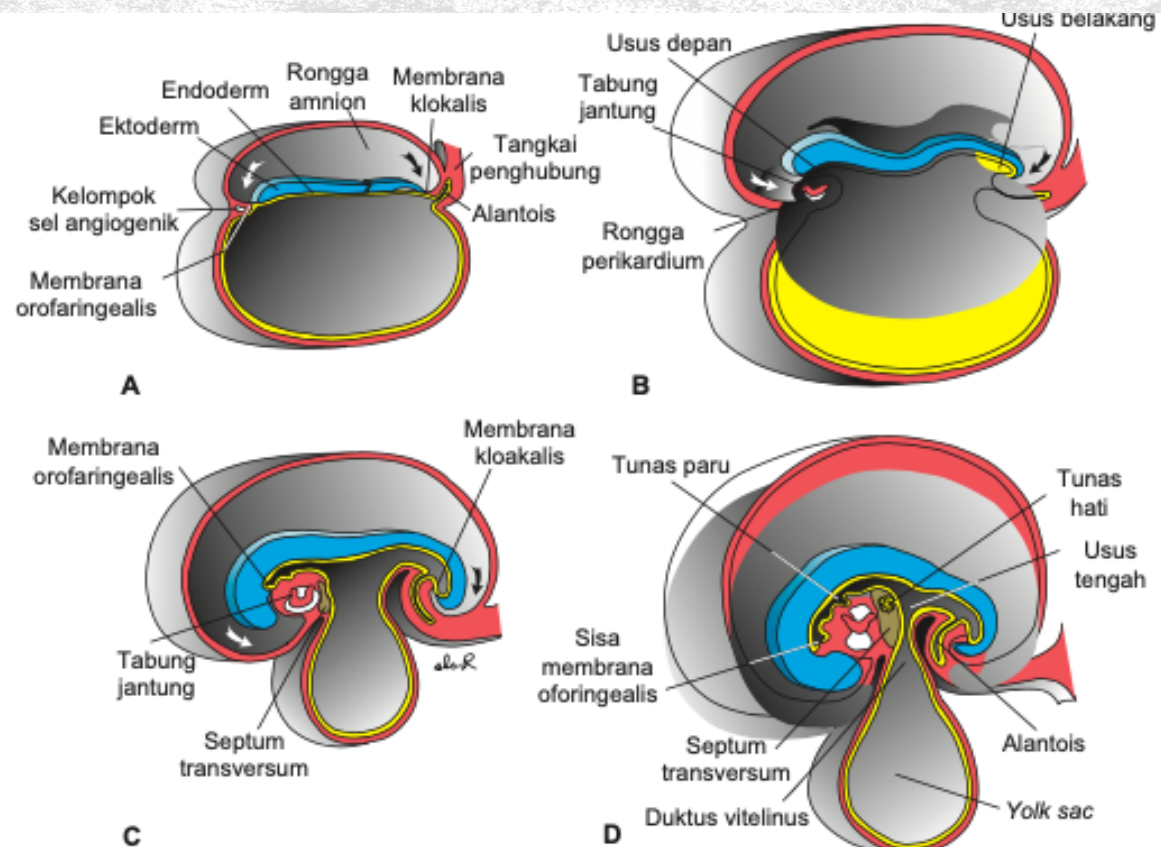
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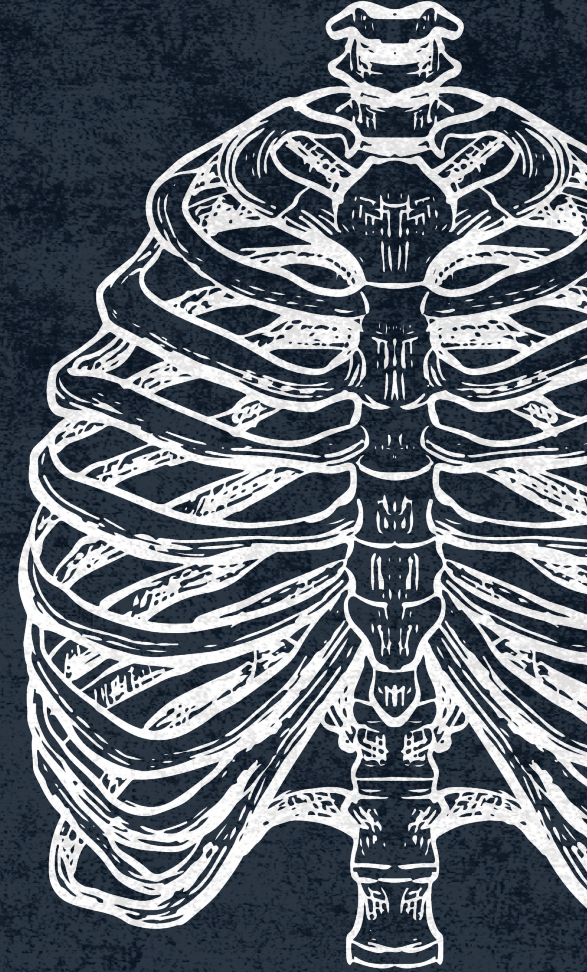


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LATERALIZATION



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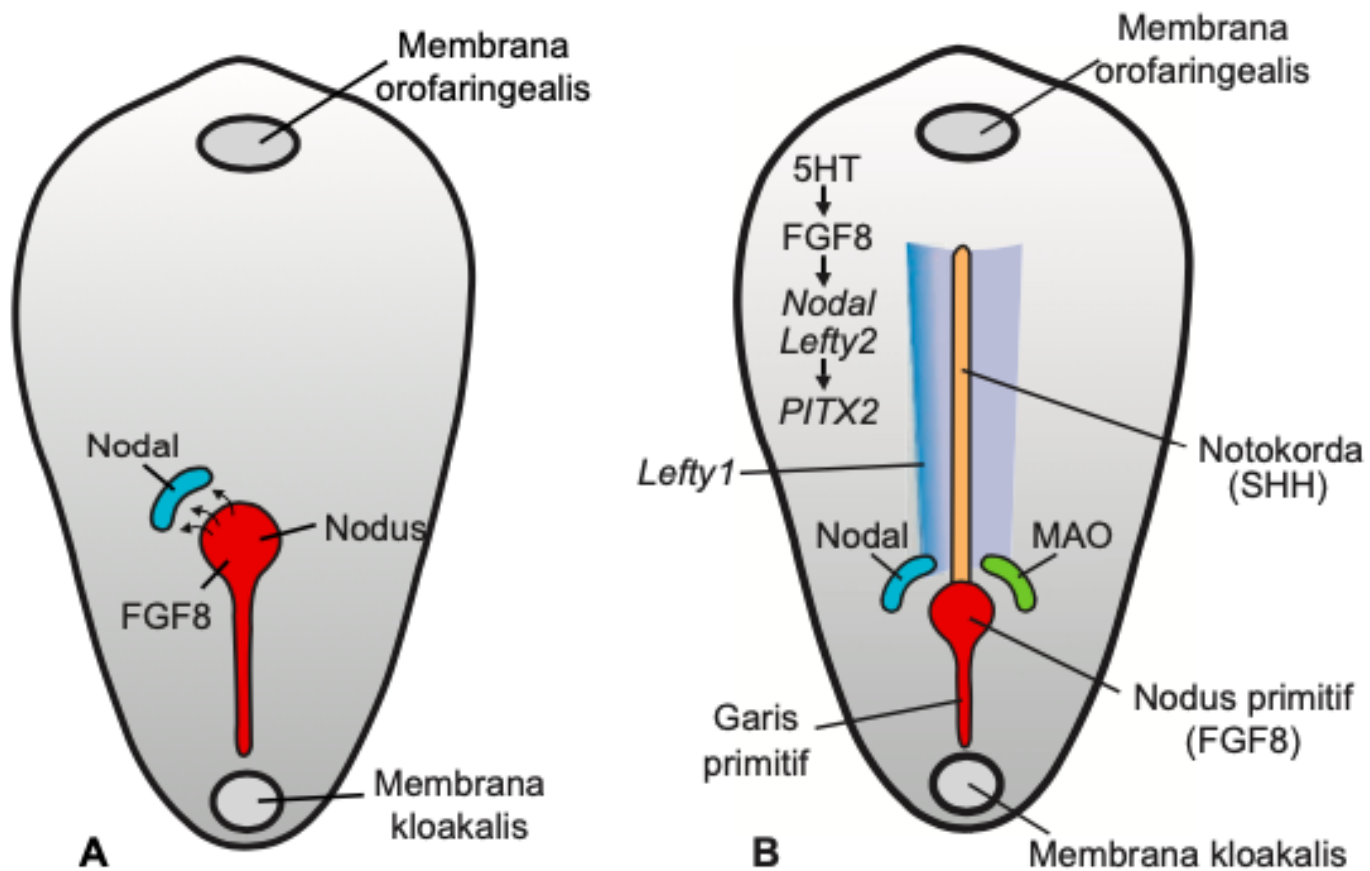
LATERALIZATION

Now, there is some serious question, why heart is formed in left side of chest? Why gaster is formed in epigastric sinistra region?





LATERALIZATION



Pregnant woman is
prohibited to consume SSRI
(Anti depressant)!



REFERENCES

- Crozet N. 1994. Réaction acrosomique et fécondation [Acrosome reaction and fertilization]. Contraception, fertilité, sexualité (1992), 22(5), 328–330.
- Moore, Keith L, Persaud, T.V.N. 2013. The Developing Human Clinically Oriented Embryology (Ed. 9). Philadelphia: Elsevier Saunders.
- Ramesh, T., Nagula, S. V., Tardieu, G. G., Saker, E., Shoja, M., Loukas, M., Oskouian, R. J., & Tubbs, R. S. 2017. Update on the Notochord Including its Embryology, Molecular Development, and Pathology: A Primer for the Clinician. Cureus, 9(4), e1137. <https://doi.org/10.7759/cureus.1137>
- Sadler, T.W. 2019. Langman's medical embryology, 14th ed. . Philadelphia: Wolters Kluwer.





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